reaction-box

The Reaction Box

A locally symmetric cellular generator, the property that makes it information-preserving, and the question of which other algorithms share that property.

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1. Motivation

The Coylean map is generated by tiling a single local rule — the **reaction box** — across a dyadic grid. The rule has a striking property: its four-port relation is symmetric under interchange of input and output ports. The same rule that propagates state forward also recovers state backward.

This is what gives the resulting maps their **conservation of information**: the entire interior of a patch can be reconstructed from its two boundary seams (and vice versa), because the box that built the patch is the same box that inverts it.

This document has two goals.

1. **Define the reaction box precisely** and exhibit the symmetry as a concrete involution on its port-state space.
2. **Frame the search for other rules with this symmetry**, including concrete measures that quantify how close a given local rule is to having it.

The hope is that the symmetry is not a one-off curiosity of the Coylean algorithm but the local signature of a broader class of information-preserving cellular generators — and that we can recognize members of that class systematically.

2. The Reaction Box

A reaction box is a four-port combinational element with one scalar parameter.

```
                N (input: v)
                     │
                     ▼
               ┌───────────┐
               │           │
W (input:h)    │     P     │ E (output: r)
        ──────►│           │──────►
               │           │
               └───────────┘
                     │
                     ▼
                S (output: d)
```

- **Vertical input** `v ∈ {0,1}` arrives at the north port.
- **Horizontal input** `h ∈ {0,1}` arrives at the west port.
- **Priority parameter** `P ∈ {V, H}` is a scalar choosing which axis wins when both inputs are active.
- **Vertical output** `d ∈ {0,1}` leaves at the south port.
- **Horizontal output** `r ∈ {0,1}` leaves at the east port.

The box implements a deterministic function

$$f\_P : \{0,1\}^2 \to \{0,1\}^2, \quad f\_P(v, h) = (d, r).$$

The convention to keep in mind: each axis has an input port and a complementary output port on the opposite side (N ↔ S vertical, W ↔ E horizontal).

3. The Truth Tables

The Coylean reaction box is defined by the following two tables (one per priority setting). They are derived from `reactionFromPriority(vertical, horizontal, …)` in `coylean-core.js`.

Priority V — vertical wins ties (`downWins = true`):

v	h	d	r
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

Priority H — horizontal wins ties (`downWins = false`):

v	h	d	r
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	1

Two observations to set up the rest of the paper:

1. Within each table, the four output rows are a permutation of the four possible 2-bit outputs. The map `f_P` is a **bijection** for each priority.
2. The H table is obtained from the V table by swapping `v ↔ h` (and correspondingly `d ↔ r`). The two priority settings are themselves dual under axis exchange.

4. The Symmetry Property

The reaction box is symmetric in a specific, strong sense: **interchanging complementary inputs with their outputs leaves the relation invariant.** Formally, define the port-swap involution

```
σ : (v, h, d, r) ↦ (d, r, v, h).
```

Let `R_P ⊆ {0,1}⁴` be the relation `{ (v,h,d,r) : f_P(v,h) = (d,r) }`. Then for both priorities,

```
σ(R_P) = R_P.
```

Equivalently: applying the rule twice returns to the starting state. For each priority `P`,

```
f_P ∘ f_P = id    (the box is an involution).
```

This can be read directly off the tables above. Re-feeding `(d, r)` back in as `(v, h)` recovers the original `(v, h)`:

```
Forward                      Reverse (= same rule)
─────────────────────        ─────────────────────

      v                            d
      │                            ▲
      ▼                            │
   ┌───────┐                   ┌───────┐
h →│   P   │→ r                │   P   │
   └───────┘  becomes inputs ⇄ └───────┘ ← inputs (r, d)
      │                            ▲       outputs (v, h)
      ▼                            │
      d                            v
```

This is the precise meaning of "interchanging complementary inputs and outputs produces the same result." The box has no preferred direction of flow — it is a self-inverse logical gate, not a one-way circuit. The forward sweep that turns boundary seams into an interior, and the backward sweep that recovers seams from the interior, run **the same program in opposite orientations**.

5. From the Box to the Map

Cellular structure. The map is obtained by tiling the reaction box across a 2D grid:

```
     column index i →

   ┌───┬───┬───┬───┬───┐
   │ P │ P │ P │ P │ P │     • each cell holds one reaction box
   ├───┼───┼───┼───┼───┤     • d flows downward (column-wise)
   │ P │ P │ P │ P │ P │     • r flows rightward (row-wise)
   ├───┼───┼───┼───┼───┤     • output of one cell = input of next
   │ P │ P │ P │ P │ P │       (south output → next cell's north
   ├───┼───┼───┼───┼───┤        input; east output → next cell's
   │ P │ P │ P │ P │ P │        west input)
   └───┴───┴───┴───┴───┘

     row index j ↓
```

The priority `P` at each cell is **not constant** — it is set per cell by a function of the absolute coordinates `(i, j)`. In the Coylean case this is the 2-adic valuation: comparing `ν₂(i + hInitCol)` with `ν₂(j + vInitRow)` selects V or H at that cell. Because the priority depends on the **absolute** indices, the tiling is *not* translation-invariant — the map has a preferred origin.

What is needed to talk about "the map":

Scope	Information required
One cell	the reaction box rule + a priority bit
Finite patch	the rule + per-cell priorities (i.e. anchor position) + two boundary seams
Infinite map	the rule + a generative seam rule + an anchor

The patch-level conservation of information is a direct consequence of the box-level involution: stitching involutions side-by-side along shared ports yields a globally invertible relation between boundary and interior. The seam rule (e.g. all-ones, impulse, periodic, universe-seed) is what extends this from finite patches to a well-defined infinite object.

6. The Family of Symmetric Local Rules

Take the reaction box's structural skeleton and treat its choices as parameters. We get a candidate family of "Coylean-like" local rules:

- **Port arity.** A box with `n` input ports and `n` complementary output ports. `n = 2` is the case above; `n = 3` would be a 3D lattice (NWB → SEF), and so on.
- **Local state space.** `{0,1}` for the Coylean box. Could be any finite alphabet `Σ`. A reversible Σ-valued rule with the port-swap symmetry is the natural generalization.
- **Priority parameter.** Any finite indexing set `Π` whose values select among "tie-breaking" sub-rules. `Π = {V, H}` for Coylean.
- **Priority assignment.** A function `Z² → Π` (or `Zⁿ → Π` more generally) chosen per cell. For Coylean this is 2-adic-valuation comparison; other arithmetic structures (3-adic, mixed-radix, recursive block patterns) give other families.
- **Boolean / algebraic rule.** What the box actually computes — for Coylean, XOR-based with priority breaking ties. Other group-valued rules (mod-3 addition, permutation groups, etc.) over the chosen alphabet are possible.

The interesting question is which combinations preserve the **port-swap involution property**. A rule that is merely bijective is already a reversible cellular automaton (the Margolus / Toffoli line of work). A rule that is bijective *and* a port-swap involution is more restrictive: it commits to the same physical mechanism running forward and backward. This is the property to chase.

Candidate sources to mine:

- Reversible cellular automata literature (Margolus neighborhoods, Toffoli gates, block CA).
- Lattice gas automata; in particular Margolus-style discrete fluid rules can be checked for self-inverse local rules.
- Domino-tiling and dimer-model transfer matrices, where local rearrangement moves are often involutive.
- Tropical / min-plus semirings: priority-style tie-breaking is natural there.
- Quantum reversible circuits: any rule realizable by a Hadamard-free Clifford gate over the computational basis projects to a classical

involution.

A box-level involution is a small object — at most a `(2^n)! / 2`-sized set of candidate rules for `n` ports over `{0,1}`. Enumerating them exhaustively for small `n` and asking which support interesting priority-driven tilings looks tractable.

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7. Measuring Symmetry

Given an arbitrary local rule, how do we judge whether it has the Coylean property? Five measures, increasing in informativeness:

7.1 Bijectivity (necessary condition)

`f_P` must be a bijection from Σ^n (inputs) to Σ^n (outputs) for each priority `P`. Equivalently, the rule's truth table, viewed as a function, has no repeated outputs. This is a yes/no test.

7.2 Port-swap involution (the property itself)

`f_P ∘ f_P = id` for every `P`. Equivalently, the relation `R_P` on `(inputs, outputs)` is invariant under input/output exchange. Yes/no.

7.3 Hamming distance to nearest involution

If a rule fails the involution test, **how far off is it?** Define

```
d_inv(f_P) = min over involutions g of |{ x : f_P(x) ≠ g(x) }|
```

— the minimum number of truth-table cells we would have to edit to make `f_P` an involution. This gives a graded score; `d_inv = 0` is exactly the Coylean property.

7.4 Information-theoretic capacity

Treat the inputs as random variables, run them through the rule, and measure the mutual information between input and output:

```
I(f_P) = H(inputs) − H(inputs | outputs).
```

A perfect involution achieves `I = H(inputs)` — every bit of input is recoverable from the output. Lossy rules score lower. This generalizes gracefully to probabilistic or stochastic rules.

7.5 Tile-extension test

The local property is only interesting if it lifts to the tiled lattice. Take a candidate rule, tile it across an `x · y` patch with some priority assignment, and check whether the interior arrows determine the boundary seams. Compute

```
rank(interior → boundary)
```

over an appropriate algebra. Full rank means the patch is in bijection with `(seams + anchor)` — i.e. conservation of information lifts to the patch level. Rank deficit gives a precise measure of information loss in the cellular extension.

Measures 7.1–7.3 are properties of the local rule alone. Measure 7.4 is a probabilistic relaxation. Measure 7.5 tests whether the local property survives tiling — which is the property we ultimately care about.

8. Open Questions

- **Classification.** For port arity `n = 2` over `Σ = {0,1}` and priority set `Π = {V, H}`, how many involutive rules are there? Up to obvious symmetries? Which of them give rise to non-trivial tilings (i.e. priority assignments that produce structured maps rather than periodic / trivial ones)?
- **Beyond 2-adic priorities.** The 2-adic valuation gives the Coylean map its self-similar dyadic structure. Replacing it with a 3-adic valuation, or with a non-arithmetic recursive partition, should produce structurally different but still information-preserving maps. What do those look like?
- **Higher arity.** A 3D reaction box (3 inputs, 3 outputs, port-swap involution) tiled on the integer lattice would produce a "Coylean cube." Does the involution property survive in 3D? What priority structures are natural there?
- **Continuum limits.** Is there a sensible continuum limit of the Coylean tiling under dyadic-zoom rescaling that preserves the information-conservation property? Differential operators that are formally self-adjoint under boundary-pairing are the analog target.
- **Connection to reversible computing.** The Coylean box is a reversible logic gate with a specific symmetry. Where does it sit in the standard taxonomy of reversible gates (Toffoli, Fredkin, etc.)? Are there universal gate sets built from rules with this stronger property?

9. Summary

The Coylean map is fully determined by a local rule — the reaction box — that is **a port-swap involution**: feeding outputs back in as inputs recovers the original inputs. This single property is responsible for the map's conservation of information at every scale: from one cell, to an `x · y` patch, to the full infinite map (given a generative seam rule and an anchor).

The interesting research direction is no longer "what does this specific map look like" but "which local rules share this property, and how do we recognize them." Sections 6 and 7 give the parameter space and the measures; the question is what lives in there besides the Coylean rule.